We start field theory from here, an important theorem is the main theorem in Algebra 2:

Theorem 1 (Main Theorem)**.** *Let*

- K *be a field*
- *Non constant term* $f(x) = a_n x^n + \ldots + a_1 x + a_0 \in K[x]$ *an irreducible, with* $a_i \in K$
- *Given the usual projection* $\pi : K[x] \to K[x]/(f), \pi(p) = p + (f)$*, if we write* $\pi(x) = x + (f) = \alpha$*, then:*

1. $L = K[x]/(f)$ *is a field.*
2. $f(\alpha) = 0$*, that is,* α *is a root of* f *in* L*.*
3. $[L : K] = \deg f = n$ *and* $1, \alpha, \ldots, \alpha^{n-1}$ *is a basis for* L/K*.*
4. *If* F/K *is a field extension, and* $f(\beta) = 0$ *for some* $\beta \in F$*, then* $K \hookrightarrow F$ *extends to a unique homomorphism* $\Psi : L \hookrightarrow F$ *such that* $\Psi(\alpha) = \beta$*. The image* $\Psi(L)$ *is* $K(\beta)$*, the smallest subfield of* F *containing* K *and* β*.*

Other facts about finite fields are also relevant here.

1 Separable and Normal Extensions

From here on we always work on F, a field. The minimal polynomial (over F) of $\alpha \in E$, where E/F is an extension, is denoted as $m_F(\alpha)$. From here on, E/F is always an algebraic extension.

Definition 1. A polynomial $f \in F[x]$ is **separable over** F if, in its irreducible factorization, all the components don't have repeated roots (no "powers" in the factorization in $\overline{F}$).

Definition 2. Given an extension E/F, an element $\alpha \in E$ is **separable over** F if $m_F(\alpha) \in F[x]$ is separable over F. If all such αs are separable, E is called **separable over** F.

Theorem 2 (Equivalent Definition for Separability)**.** *If* $f \in F[x]$ *is irreducible:*

- f *is separable* $\Leftrightarrow \gcd(f, f')$*, evaluated in* $F[x]$*, is not constant*
- *If* $\mathrm{Char}(F) = 0$*, then every (irreducible) polynomial is separable.*
- *If* $\mathrm{Char}(F) = p$*, a prime, then* f *is inseparable* $\Leftrightarrow f' \equiv 0 \Leftrightarrow f$ *is in the form of* $g(x^p)$ *for some* $g \in F[x]$*.*

All elements in $\mathbb{Q}[x], \mathbb{F}_q[x]$, where q is prime powers, is always separable. So any field extension base on it is always separable.
An inseparable polynomial example is $x^p - t$ in the extension $\mathbb{F}_p(t)(\sqrt[p]{t})/\mathbb{F}_p(t)$. So not all splitting fields are separable.

Theorem 3 (Primitive Element Theorem)**.** *If* $[E : F] < \infty$ *and* E/F *separable, then there is* $\alpha \in E$ *such that* $E = F(\alpha)$*.*

Theorem 4. *Separability is transitive. That is, if* $E/K/F$ *is tower of extension,* E/F *separable* $\Leftrightarrow E/K, K/F$ *separable.*

We are interested in automorphisms σ, fixing/respecting the structure of F.

Theorem 5 (Lifting Lemma)**.** *Given field homomorphism* $\sigma : F \to \overline{F}$*, there is always* $\widehat{\sigma} : E \to \overline{F}$ *such that* $\widehat{\sigma}|_F = \sigma$*.*

- *Moreover, if* E/F *is a finite extension, if* E/F *is separable, then there is exactly* $[E : F] := n$ *of such distinct* $\widehat{\sigma}$*. Otherwise, it has no more than* n *such* $\widehat{\sigma}$*.*

Following the theorem, we are most interested when $\sigma = \mathrm{id}$ in F. Given an homomorphism $\sigma : E \to E$,

Definition 3. We call such σ a F**-homomorphism**. If for all F-homomorphism σ we have $\sigma(E) = E$, then we say E/F is **normal**.

Theorem 6 (Equivalent Definition of Normacy)**.** *If* E/F *is a finite extension, the following are equivalent:*

1. E/F *is normal over* F*.*
2. *For all the irreducible polynomial* $f \in F[x]$*, if* $\alpha \in E$ *is as root of* f*, then all the roots of* f *(considered as a factorization in* $\overline{F}$*) are in* E*.*
3. E *is the splitting field for some set of polynomial* S *over* F*.*

Note that if E/F *is not finite,* $(1) \Leftrightarrow (2)$ *still holds.*

2 Some Galois Theory

Definition 4. A normal and separable extension is called **Galois**. The collection $\{\sigma : \sigma|_F = \mathrm{id}\}$, where σ comes from the lifting lemma, is called the **Galois Group**, denoted by $\mathrm{Gal}(E/F)$.

Key examples of splitting field over $\mathbb{Q}$ are $x^3 - 2$ (roots are $\sqrt[3]{2}$, $\sqrt[3]{2}\omega$, and $\sqrt[3]{2}\omega^2$, where ω is the cube root of unity) and $8x^3 + 4x^2 - 4x - 1$ (roots are $\cos\left(\frac{2k\pi}{7}\right), k = 1, 2, 3$). They have degrees of 6 and 3 respectively. See Section 6.3 on Ash for a detailed calculation, their Galois Group is S_3 and $\mathbb{Z}/3\mathbb{Z}$ respectively.

Definition 5. Let $G = \mathrm{Gal}(E/F)$ be the Galois group of the extension E/F. If H is a subgroup of G, the **fixed field** of H is

$$\mathcal{F}(H) = \{x \in E : \sigma(x) = x \text{ for every } \sigma \in H\}.$$

If $F \leq K \leq E$, define the "**fixed group**" of K to be

$$\mathcal{G}(K) = \mathrm{Gal}(E/K) = \{\sigma \in G : \sigma(x) = x \text{ for every } x \in K\}$$

(we used quotes because the name is not standard.)

Theorem 7 (Dedekind)**.** *Distinct "characters" (for now, the homomorphism in Galois Group* $\mathrm{Gal}(E/F)$*) are linearly independent over* E*.*

The main catch of the theory is that $\mathcal{F}, \mathcal{G}$ are contravariant functors between the category **Grp** and **Fld**. It preserves the normalcy of groups as well. Here is a verbose version:

Theorem 8 (Galois Correspondence)**.** *Let* $F \leq E \leq K$ *be a tower of fields,* E/F *Galois, then:*

- $\mathcal{F}$ *is a bijective map from subgroups to intermediate fields, with inverse* $\mathcal{G}$*.*
- *(Reverse Inclusion.) If* $H_1 \leq H_2$ *then* $\mathcal{F}(H_1) \geq \mathcal{F}(H_2)$*, and if* $K_1 \leq K_2$*, then* $\mathcal{G}(K_1) \geq \mathcal{G}(K_2)$*.*
- *Suppose that the intermediate field* K *corresponds to the subgroup* H *under the Galois correspondence. i.e.* $\mathrm{Gal}(E/K) = H$*, i.e.* $\mathcal{F}(H) = K$*, Then*
  - E/K *is always a normal field extension (hence Galois);*
  - $[K : F] = [G : H]$ *and* $[E : K] = |H|$
  - *For any* $\sigma \in \mathrm{Gal}(E/F)$*, then the field* $\sigma K = \{\sigma(x) : x \in K\}$ *corresponds to the conjugate subgroup* $\sigma H \sigma^{-1}$*. i.e.* $\mathrm{Gal}(E/\sigma K) \cong \sigma H \sigma^{-1}$*.*
  - K/F *is a normal field extension* $\Leftrightarrow$ H *is a normal subgroup of* G*, and* $\mathrm{Gal}(K/F) \cong G/H$*.*

Definition 6. Now we move on to a specific type of extension.

- A **cyclotomic extension** E/F is a field extension where E is the splitting field of the polynomial $x^n - 1$.
- The roots of $x^n - 1$ in $\overline{F}$ are called the **roots of unity**, all the roots of unity form a $\leq n$-element, NOT NECESSARILY EQUAL cyclic subgroup of E^*.
- A **primitive** n^{th} **root of unity**, if it exists, is a generator of the above subgroup, which is of order n.

It is obvious that if a primitive n^{th} root of unity does exist, call it ω_n, then $E = F(\omega_n)$.
The non-existence of primitive roots can be demonstrated by the example of $x^{20} - 1 = (x^4 - 1)^5$ over $\mathbb{F}_5$.

Theorem 9. *Primitive* n^{th} *roots exists* $\Leftrightarrow \gcd(\mathrm{char}\, F, n) = 1$.

Theorem 10. *If* ω_n *is a primitive root of unity over* $\mathbb{Q}$*, then* $\mathrm{Gal}(\mathbb{Q}(\omega_n)/\mathbb{Q}) = U_n$*, the multiplicative group of* $\phi(n)$ *elements.*

The proof of the theorem introduces the following powerful polynomial:

Definition 7. The nth **cyclotomic polynomial**, ψ_n, is

$$\psi_n(x) := \prod_i (x - \omega_i)$$

where $\omega_1, \omega_2 \cdots$ are n^{th} primitive roots of unity, so there is exactly $\phi(n)$ of them. (why, *a priori*?)

Theorem 11. *The cyclotomic polynomial is actually irreducible over* $\mathbb{Q}$ *and has integer coefficients. Moreover,*

$$x^n - 1 = \prod_{d|n} \psi_d(x).$$

3 Algebraic Number Theory

In this section and the next, we assume R is a commutative ring with unity.

Definition 8. Let $A \subseteq R$, A a ring.

- $x \in R$ is **integral over** A if x is a root of some polynomial with coefficient in A. If for all $x \in R$ this happens, then we say R is **integral over** A.
- In particular, if $R = \mathbb{C}$, then x is called an **algebraic integer** if x is integral over $\mathbb{Z}$.
- The set $\{x \in R : x \text{ is integral over } A\}$ is called the **integral closure of** A **over** R, denoted by A_c.
- A is **integrally closed over** R if $A = A_c$.

Now a confusion definition which is almost always used:

Definition 9. A is **integrally closed** if A is an integral domain and integrally closed over $\mathrm{Frac}(A)$.

Definition 10. A R-module S is said to be **faithful** if its **annihilator** $\mathrm{Ann}_R(S) = \{r \in R \mid s \in S \Rightarrow rs = 0\} = 0$.

Theorem 12. *A_c is a ring itself, we can show this by considering a list of equivalent conditions:*

- *x is integral over A;*
- *The A-module $A[x]$ is finitely generated;*
- *x belongs to a subring B of R such that $A \subseteq B$ and B is a finitely generated A-module.*
- *There is a subring B of R such that B is a finitely generated A-module and $xB \subseteq B$.*
  - *If R is a field, the assumption that B is a subring can be dropped (as long as $B \neq 0$).*
- *There is a faithful $A[x]$-module B that is finitely generated as an A-module.*

Theorem 13. *If A is integral over B, then A is a field $\Leftrightarrow$ B is a field.*

Definition 11. This is what we will consider from now on:
A is an integral domain, $K = \mathrm{Frac}(A)$, L is a finite separable extension of K, B is the integral closure of B over L. We will call this our **AKLB setup** for now, but the most important setting is letting $A = \mathbb{Z}$: (the arrow is inclusion, the dotted arrow means we are actually *defining* B)

$$\begin{array}{ccc} L & \xleftarrow{\quad\quad} & B \\ \uparrow & & \\ K = \mathrm{Frac}(A) & \longleftarrow & A \end{array} \qquad \begin{array}{ccc} L & \xleftarrow{\quad} & \mathcal{O}_L \\ \uparrow & & \\ \mathbb{Q} & \longleftarrow & \mathbb{Z} \end{array}$$

As $A = \mathbb{Z}$, L above is called a **number field** we called B the **ring of integers** of L and denote it by $\mathcal{O}_L$.

Theorem 14. *If $A = A_c$, then A is noetherian as a ring $\Rightarrow$ B is noetherian as a ring.*

Now let's identify some ring of integers:

Theorem 15. *If d is square-free, $L = \mathbb{Q}(\sqrt{d})$, then If $d \equiv 1$ mod 4, then $\mathcal{O}_L = \mathbb{Z}[\sqrt{d}]$, otherwise, $d \equiv 3$ mod 4, then $\mathcal{O}_L = \mathbb{Z}\left[\frac{1+\sqrt{d}}{2}\right]$*

Now let's look at some important "numbers": given a (number field / AKLB setup) L, any $x \in E$ gives a map

$$\mathcal{M}_x : E \to E \qquad \mathcal{M}_x(a) = ax$$

This is a K-linear map, and we can do whatever linear algebra we want. However, we can do better, with our discussion of Galois Theory above, we can produce a lot of alternative definitions for the same object:

Theorem 16. *Assuming the AKLB setup, let* $[L:K]=n, [K(x):K]=d, x_1, \cdots x_d$ *are the roots of the minimal polynomial* $m_F(x)$*, by lifting lemma, there are* n F*-homomorphisms* $\sigma_1, \cdots \sigma_n : L \to \overline{L}$*.*

Then, given $x \in L$*, the following three expressions are the same:*

$$\textit{Trace of } \mathcal{M}_x, \qquad \frac{n}{d}\sum_{i=1}^{d} x_i, \qquad \sum_{i=1}^{n}\sigma_i(x).$$

Also, the following three expressions are the same:

$$\textit{Determinant of } \mathcal{M}_x, \qquad \left(\prod_{i=1}^{d} x_i\right)^{\frac{n}{d}}, \qquad \prod_{i=1}^{d}\sigma_i(x)$$

Also, the following characteristic polynomial and the minimal polynomial (in y*) raised to the power* n/d *are the same:*

$$\det(yI - \mathcal{M}_x) = (m_F(x))^{n/d}.$$

Note:

- The trace and determinant actually require a choice of basis in L/K, but since for any choice we will get the same trace and determinant, it is indeed well-defined.
- Moreover, from the first expression it is easy to see that for all $x \in L, \mathrm{Tr}_{L/K}(x), \mathrm{N}_{L/K}(x) \in K$.

Definition 12. The above three quantities are called the **trace, norm** and **characteristics polynomial** of $x \in L$ of the field extension L/K, it is denoted as $\mathrm{Tr}_{L/K}(x), \mathrm{N}_{L/K}(x), \mathrm{char}_{L/K}(x)$ respectively.

Theorem 17. *The coefficient of* $m_F(x)$ *is integral over* A*, hence so is the coefficient of the characteristic polynomial, (hence) the trace, and the norm.*

The key is that if x is integral over A, so are its conjugates.

We can use the "exactly n/d extension" and proof that:

Theorem 18. *Let* $E/K/F$ *be a tower of field extensions, then* $\mathrm{N}_{E/F} = \mathrm{N}_{E/K} \circ \mathrm{N}_{K/F}, \mathrm{Tr}_{E/F} = \mathrm{Tr}_{E/K} \circ \mathrm{Tr}_{K/F}$*.*

Now a very useful result from Dedekind's Lemma:

Theorem 19. *The trace is not identically zero.*

Definition 13. Given the AKLB setup, and $\mathbf{x} = (x_1, \cdots x_n) \in L^n$, then define the **discriminant** of $\mathbf{x}$ as:

$$\mathrm{Disc}(x) := \det\left(\mathrm{Tr}_{L/K}(x_i x_j)\right)$$

Here is an example of the determinant, it is the celebrated Vandermonde matrix:

Theorem 20. *If* $x \in \overline{K}$ *and take* $L = K(x)$*, then set* $\mathbf{x} = (1, x, x^2\cdot, x^{n-1})$*. Now let* $x_1, x_2, \cdots x_n$ *be conjugates of* x*:*

$$\mathrm{Disc}(\mathbf{x}) = \left(\prod_{i<j}(x_i - x_j)\right)^2.$$

Theorem 21 (Properties of discriminant)**.** *Let* $\sigma_1 \cdots \sigma_n$ *be the* n K*-homomorphisms, again. If* $\mathbf{x} \in L^n$*, then:*

- *If* $\mathbf{y} = C\mathbf{x}$ *then* $\mathrm{Disc}(\mathbf{y}) = (\det C)^2 \mathrm{Disc}(\mathbf{x})$.
- $\mathrm{Disc}(\mathbf{x}) = (\det(\sigma_i(x_j)))^2$.
- $\{x_1, \cdots x_n\}$ *is a basis for* $L/K \Leftrightarrow \mathrm{Disc}(\mathbf{x}) \neq 0$.

Now two random theorem about AKLB setup:

Theorem 22. *If* A *is a PID then* B *is a free* A*-module rank* n*.*

Theorem 23. *If* A *is integrally closed then* A *is Noetherian implies* B *is Noetherian.*

4 Algebraic Geometry

We fix a field k and we consider the affine n-space (space of n-tuples in k) $\mathbb{A}^n$, we want to make a connection with ideals of $k[x_1, \cdots x_n]$ and "points" on $\mathbb{A}^n$

Definition 14. A **(affine) variety** is the zero set (in $\mathbb{A}^n$) of some set $S \subset k[x_1, \cdots x_n]$. We also denote

$$V(S) := \{x \in \mathbb{A}^n : f(x) = 0 \text{ for all } f \in S\}.$$

Here is a key property, any set $S \subset k[x_1, \cdots x_n]$ can be expanded to an ideal

$$(S) = \left\{ \text{finite sums in terms of } \sum_{i=1}^{n} g_i f_i : g_i \in k[x_1, \cdots x_n], f_i \in S \right\}$$

and we can check that $V(S) = V((S))$. So we will just write $V(S)$ from now on.

Definition 15. The **Zariski Topology** is the topology on $\mathbb{A}^n$ by declaring varieties as closed sets.

The above is actually a theorem, essentially checking all the axioms.

Corollary. *Any variety is a finite intersection of hyperplanes ($V(\{f\})$, where f is a single polynomial / principal ideal)*

So at least any ideals (S) can be put into a variety (a subset in $\mathbb{A}^n$), can we do the opposite?

Definition 16. The **ideal** of a subset $X \subset \mathbb{A}^n$ is the zero polynomials that vanishes on X, that is:

$$I(X) = \{f \in k[x_1, \cdots x_n] : f(x) = 0 \text{ for all } x \in X\}$$

That is indeed an ideal in $k[x_1, \cdots x_n]$, and we *almost* have our equivalence:

Theorem 24. *Here are some Properties of V, I:*

- *V, I are reverse included. I.e. $S \subseteq T \Rightarrow V(S) \supseteq V(T)$, similarly for I. In particular, $I(\emptyset) = k[x_1, \cdots x_n]$.*
- *$VIV(S) = V(S), IVI(X) = X$*
- *(First Correspondence.) If $x = (a_1, \cdots a_n) \in \mathbb{A}^n$, then $I(\{x\}) = (x_1 - a_1, \cdots x_n - a_n)$, and this ideal is maximal.*

To actually get equivalence, we need $\sqrt{J}$ instead of $J \lhd k[x_1, \cdots x_n]$. We will get to this later, let's look an important lemma

Definition 17. A **finitely generated k-algebra**, denoted by $\mathcal{A}$, is isomorphic to $k[x_1, \cdots x_n]/J$ for some ideal J and $n \in \mathbb{N}$.

Lemma (Noether Normalization)**.** *If $\mathcal{A}$ is a finitely generated k-algebra, then there are $y_1, \cdots y_r$ that are algebraically independent and $\mathcal{A}$ is integral over $k[x_1, \cdots x_n]$.*

The proof lies on this important lemma, which is a trick to remember?

Lemma. *Given a polynomial $f \in k[x_1, \cdots x_n]$, There is always a "coordinate change", an automorphism of k-algebras ϕ such that such that*

$$\phi(f) = cx_n^m + \text{ lower order terms}$$

arranged in descending powers of x_n, and $c \in k$.

Theorem 25 (Zariski Lemma)**.** *If $\mathcal{A}$ is a finitely generated k-algebra, and additionally if $\mathcal{A}$ is a field, then $[A; k] < \infty$,*

Now we can introduce the main theorem: Hilbert's Nustellensatz.

Definition 18. The radical of any ideal $I \lhd R$, denoted by $\sqrt{I}$:

$$\sqrt{I} := \{x \in R : x^r \in I \text{ for some } r \geq 1\} = \bigcap_{I \subseteq P, P \text{ prime}} P$$

Theorem 26. *Consider the k-algebra $\mathcal{A} := k[X_1, \cdots X_n]$, where the X_i, this time, are not necessarily algebraically independent, let $f_1, \cdots f_r \in \mathcal{A}$*

Let J be an ideal of $k[x_1, \cdots x_n]$, where this time the variables are algebraically independent then if k is algebraically closed, the following are equivalent:

1. *Maximal ideals of $k[x_1, \cdots x_n]$ must be in the form of $(x_1 - \lambda_1, x_2 - \lambda_2 \cdots, x_n - \lambda_n)$, for $\lambda_i \in K$.*
2. *(Weak Nullstellensatz) $(f_1, \cdots f_r) = (1) \Leftrightarrow V(\{f_1, \cdots f_r\}) = \emptyset$, or just $V(J) = \emptyset \Leftrightarrow J = (1)$*
3. *(Strong Nullstellensatz) $g \in \sqrt{(f_1, \cdots f_r)} \Leftrightarrow V(\{f_1, \cdots f_r\}) \subseteq V(g)$, or $IV(J) = \sqrt{J}$*